INCH-POUND

MIL-S-44441
30 September 1992

MILITARY SPECIFICATION

SQUASH, BUTTERED, POWDER

(FOR ORAL LIQUID FEEDING)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers buttered squash powder, unit packed in pouches or cans, for oral liquid feeding of hospitalized patients. This product is one component of a meal that, when consumed with other components in three meals or supplementary nourishments, provides a nutritionally adequate, orally consumed diet (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5018, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 8940

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ment by blowing it out with moisture-free compressed air Clean control equip

SPECIFICATIONS

FEDERAL

A-A-20032 = Potatoes, White, Dehydrated

MILITARY

MIL-P-44151 - Packaging into Pouches and Meal Assembly of Oral Liquid Feeding Powders

MIL-M-44458 - Meal Module, 10-Patient, Packaging of Oral Liquid Feeding Powders into Cans and Assembly of

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

U.S. DEPARTMENT OF AGRICULTURE (USDA)

U.S. Standards for Grades of Butter

U.S. Standards for Instant Nonfat Dry Milk

General Specifications for Approved Dairy Plants and Standards for Grades of Dairy Products (7 CFR Part 58)

(Copies are available from the Dairy Standardization Branch, Dairy Division, Agricultural Marketing Service, U.S. Department of Agriculture, P.O. Box 96456, Room 2750 South Building, Washington, DC 20090-6456.)

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U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES (HHS), U.S. FOOD AND DRUG ADMINISTRATION (FDA)

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder (21 CFR Parts 1-199)

(Copies are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN ASSOCIATION OF CEREAL CHEMISTS (AACC)

Approved Methods of the American Association of Cereal Chemists

(Application for copies should be addressed to the American Association of Cereal Chemists, 3340 Pilot Knob Road, St. Paul, MN 55121.)

AOAC INTERNATIONAL

Official Methods of Analysis of the AOAC

(Application for copies should be addressed to the AOAC International, 2200 Wilson Boulevard, Suite 400-CD, Arlington, VA 22201-3301.)

NATIONAL ACADEMY OF SCIENCES

Food Chemicals Codex

(Application for copies should be addressed to the National Academy Press 2101 Constitution Avenue, N.W., Washington, DC 20418.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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indicated in para. 3.3.1.1. Rollers, pressure pairs, auxiliary compression, as

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.4.

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent or insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

3.2.1 Squash powder, freeze dried. The squash powder shall be blanched and freeze-dried yellow butternut squash and 100 percent shall pass through a U.S. Standard No. 20 sieve. The squash powder shall have the color, odor, and flavor typical of butternut squash. The maximum moisture content shall not exceed 3.0 percent.

3.2.2 Butter powder. The butter powder shall be made by encapsulating salted or unsalted butter that meets or exceeds the specifications for Grade A butter and shall conform to the definition of butter as defined in the United States Standards for Grades of Butter. The butter powder shall be manufactured in a plant that meets the requirements specified in the General Specifications for Approved Dairy Plants and Standards for Grades of Dairy Products (7 CFR Part 58). The butter powder shall be a light yellow, free-flowing powder with sweet cream butter flavor. The butter powder shall contain an encapsulating agent, such as nonfat dry milk, and 73 percent fat from butter. The moisture content shall not exceed 4.0 percent. The butter powder shall be free of hard lumps and shall be of such size that 100 percent shall pass through a U.S. Standard No. 12 sieve. An antioxidant shall be added to improve stability (see 6.4.1).

3.2.3 Potato granules, instant. Instant potato granules shall meet the requirements of form (b), style (1) of A-A-20032. In addition, the instant potato granules shall have a maximum moisture of 8.0 percent and maximum reducing sugars of 3.0 percent. The potato granules shall have a minimum Vitamin C content of 36.0 milligrams per ounce.

3.2.4 Sugar, white, granulated. Sugar shall be white, refined, granulated cane or beet sugar or a combination thereof.

3.2.5 Milk, nonfat, dry, instant. The instant, nonfat, dry milk shall be U.S. Extra Grade as defined in the United States Standards for Instant Nonfat Dry Milk (7 CFR Part 58).

3.2.6 Creaming agent, non-dairy. The non-dairy creaming agent shall be a cold water dispersible creaming agent. It shall possess a white to cream color, shall be free of hard lumps, and 100 percent shall pass through a U.S. Standard No. 20 sieve. The non-dairy creaming agent shall contain 65 to 75 percent fat and a moisture content that does not exceed 4.0 percent (see 6.4.2). The non-dairy creaming agent shall be manufactured in a USDA approved plant that meets the requirements specified in the General Specifications for Approved Dairy Plants and Standards for Grades of Dairy Products (7 CFR Part 58).

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3.2.7 Salt. Salt shall be noniodized, white, refined sodium chloride, with or without anti-caking agents and shall comply with the purity standards for sodium chloride of the Food Chemicals Codex.

3.2.8 Nutmeg. Ground nutmeg shall be derived from dried kernel of Myristica fragrans Houtt. The nutmeg shall have a dull brown color. The ground nutmeg shall contain not less than 6.5 mL of volatile oil per 100 grams and be of such size that not less than 95 percent shall pass through a U.S. Standard No. 20 sieve.

3.2.9 Pepper, white, ground. Ground white pepper shall be derived from the dried mature berries of Piper nigrum L. from which the outer covering or the outer and inner covering have been removed. The ground pepper shall have a characteristic, penetrating odor, a hot biting, pungent flavor, and a light color. The ground white pepper shall contain not less than 1.0 mL of volatile oil per 100 grams of ground white pepper and be of such size that not less than 95 percent shall pass through a U.S. Standard No. 40 sieve.

3.3 Preparation and processing. Processing shall be on a continuous basis.

3.3.1 Preparation of buttered squash powder. The buttered squash powder shall be formulated and prepared as follows:

Ingredients	Percent by weight
Squash powder, freeze-dried	68.46
Butter powder	10.39
Potato granules, instant	7.00
Sugar, white, granulated	5.00
Milk, nonfat, dry, instant	4.00
Creaming agent, non-dairy	4.00
Salt	0.50
Nutmeg	0.60
Pepper, white, ground	0.05

a. All ingredients shall be combined and uniformly blended in a suitable mixer until no lumps exist, and the product is homogeneous.

b. The product shall be handled in such a manner so as to prevent an increase in moisture content (see 3.4.1) and completely packaged and sealed to meet finished product requirements.

3.3.2 Pouch filling and sealing. Each pouch shall be filled and sealed in accordance with the filling and sealing requirements specified in MIL-P-44151. Each pouch shall be filled to comply with the finished product requirements.

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3.3.3 Can filling and sealing. Each can shall be filled and sealed in accordance with the filling and sealing requirements specified in MIL-M-44458. Each can shall be filled to comply with the finished product requirements.

3.4 Finished product requirements.

3.4.1 Dry product. The dry product shall comply with the following requirements:

a. There shall be no foreign material such as, but not limited to, dirt, insect parts, hair, wood, glass, or metal.

b. There shall be no foreign odor or flavor such as, but not limited to, burnt, scorched, stale, sour, rancid, musty, or moldy.

c. There shall be no color foreign to the product.

d. The net weight of each pouch shall be not less than 23 grams and not greater than 27 grams.

e. The lot average net weight of the pouch shall be not less than 25 grams.

f. The net weight of each 401 x 602 can shall be not less than 237 grams.

g. The lot average net weight of the 401 x 602 can shall be not less than 250 grams.

h. The product shall be free from caking and lumps which do not fall apart under light pressure.

i. One hundred percent of the product shall pass through a U.S. Standard No. 7 sieve.

j. The moisture content shall not exceed 5.0 percent.

k. The aerobic plate count shall be not greater than 25,000 per gram in four of five samples and not greater than 50,000 per gram in any individual sample.

l. The Escherichia coli count shall be less than 1 per gram (no positives in the standard 3 tube MPN technique).

m. The Salmonella test shall be negative per 25 grams of product.

3.4.2 Hydrated product. Hydration of the dry powder shall be in accordance with the specified instructions on the pouch or can label, as applicable. The hydrated product shall comply with the following requirements:

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a. After hydration for oral liquid feeding, no more than 1.0 percent of the product shall remain when the product has passed through a U.S. Standard No. 20 sieve.

b. The hydrated product shall have an odor, color, and flavor that is characteristic of buttered squash.

c. For oral liquid feeding, the hydrated product shall have a consistency that allows it to be easily sipped through a drinking tube.

3.4.3 Palatability. The hydrated product shall be equal to or better than the approved preproduction sample (see 6.2) in palatability and overall appearance.

3.5 Plant qualifications. The product shall be prepared, processed and packaged in establishments meeting the requirements of Title 21, Code of Federal Regulations, Part 110, "Current Good Manufacturing Practice in Manufacturing, Packing or Holding Human Food," and the plant sanitation requirements of the appropriate Government inspection agency.

3.6 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act, and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this specification. The contractor shall ensure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and certification. Product acceptability shall be determined by the USDA. The USDA will determine the degree of inspection and supervision necessary to ensure compliance with the requirements of this specification.

4.3 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)
b. Quality conformance inspection (see 4.5)

4.4 First article inspection. When a first article is required (see 6.2), it shall be inspected in accordance with the quality assurance provisions of this specification and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this specification or any appearance or palatability failure shall be cause for rejection of the first article.

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4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.5.1 Component and material examination. In accordance with 4.1, components and materials shall be examined in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.5.1.1 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, and compliance shall be verified by examination of pertinent labels, markings, U.S. Grade Certificates, certificates of analyses, or other such valid documents acceptable to the inspection agency. If necessary, each ingredient shall be examined organoleptically or inspected according to generally recognized test methods such as the standard methods described in the Official Methods of Analysis of the AOAC International and in the Approved Methods of the American Association of Cereal Chemists, to determine conformance to the requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

4.5.2 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, filling, sealing, and packaging requirements. Any nonconformance revealed by actual examination or by review of records of formulation or of other valid documents shall be cause for rejection of the involved product.

4.5.3 Filled and sealed pouch examination. Examination of the filled and sealed pouches shall be in accordance with the quality assurance provisions of MIL-P-44151.

4.5.4 Filled and sealed can examination. Examination of the filled and sealed cans shall be in accordance with the quality assurance provisions of MIL-M-44458.

4.5.5 Net weight examination. The filled and sealed pouches or cans shall be examined for net weight in accordance with 4.5.5.1 or 4.5.5.2.

4.5.5.1 Net weight examination of pouches. The net weight of the filled and sealed pouches shall be determined by weighing each sample unit on a suitable scale tared with a representative empty pouch. Any individual net weight of less than 23 grams or greater than 27 grams shall be classified as a minor defect. Results shall be reported to the nearest gram. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be S-3 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5. In addition, the lot shall be rejected if the sample average net weight is less than 25 grams.

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4.5.5.2 Net weight examination of cans. The net weight of the filled and sealed cans shall be determined by weighing each sample unit on a suitable scale tared with a representative empty can and lid. Any individual net weight of less than 237 grams shall be classified as a minor defect. The results shall be reported to the nearest gram. The lot size shall be expressed in cans. The sample unit shall be one filled and sealed can. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 2.5. In addition, the lot shall be rejected if the sample average net weight is less than 250 grams.

4.5.6 Product examination. Eight filled and sealed pouches or cans shall be selected at random from the lot and examined for the defects listed in table I. The finding of any major defect shall be cause for rejection of the lot.

TABLE I. Product defects 1/ 2/

Category	Defect
Major	Dry product
101	Product is not free from caking and lumps which do not fall apart under light pressure
102	Any product remaining on a U.S. Standard No. 7 sieve 3/
	Hydrated product 4/ 5/
103	More than 1.0 percent of product does not pass through a U.S. Standard No. 20 sieve 6/
104	The hydrated product does not have the color, odor and flavor of buttered squash
105	The product does not have a consistency which allows it to be easily sipped through a drinking tube 7/

1/ The presence of foreign material (for example, dirt, insect parts, hair, wood, glass, metal), foreign odor or flavor (for example, burnt, scorched, rancid, stale, sour, musty, moldy), or foreign color shall be cause for rejection of the lot.

2/ Product not equal to or better than the approved preproduction sample in palatability and overall appearance shall be cause for rejection of the lot (see 3.4.3).

3/ The powder shall be poured onto a U.S. Standard No. 7 sieve and vigorously shaken.

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4/ Hydration of the dry powder from the pouch or can shall be accomplished in accordance with the pouch or can label instructions in MIL-P-44151 or MIL-M-44458, as applicable.

5/ After hydration of the required quantity from each sample unit, divide the quantity into two equal parts. Use one part for examination of defect 103 and one part for examination of defects 104 and 105.

6/ The hydrated product shall be poured onto a tared U.S. Standard No. 20 sieve. The product shall be washed for 1-1/2 minutes with slowly running, lukewarm water. The sieve shall be allowed to drain for two minutes, then excess water shall be wiped from the bottom and sides of the sieve. The remaining solids shall be weighed on the tared sieve.

7/ The product shall be easily sipped through a white or clear plastic drinking tube, 7-3/4 inches long and a 1/4 inch diameter (inside).

4.5.7 Moisture content testing. Eight filled and sealed pouches or cans shall be selected at random from the lot regardless of lot size. The pouched or canned product shall be individually tested for moisture content in accordance with the Official Methods of Analysis of the AOAC, method 945.43, with the exception that the product shall be dried for 16 hours and no sand or water added. The test results shall be reported to the nearest 0.1 percent. Any result not conforming to the requirement in 3.4.1 shall be classified as a major defect and shall be cause for rejection of the lot.

4.5.8 Microbiological testing. Five filled and sealed pouches or cans shall be selected at random from the lot regardless of lot size. The pouched or canned product shall be individually tested for microbiological levels in accordance with the Official Methods of Analysis of the AOAC, method 966.23. The diluent shall be added to each sample and allowed to stand for 15 minutes before blending the sample. Salmonella testing shall be in accordance with the Official Methods of Analysis of the AOAC, method 967.26. Any result not conforming to the microbiological requirements in 3.4.1 shall be classified as a major defect and shall be cause for rejection of the lot.

4.5.9 Packaging examination. Examination of packaging shall be in accordance with the quality assurance provisions of MIL-P-44151 or MIL-M-44458, as applicable.

5. PACKAGING

5.1 Packaging. Packaging shall be in accordance with the requirements of MIL-P-44151 or MIL-M-44458, as applicable.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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6.1 Intended use. This product is intended to be used by patients requiring an oral liquid or pureed diet (for example, those with maxillofacial injuries, wired jaws, oral surgery, dental disease, or swallowing difficulties).

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is required (see 3.1, 4.4, and 6.3).
d. Provisions for approved preproduction samples (see 3.4.3 and 6.3).
e. Type of unit pack required (see 5.1).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209-4. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Ingredient information.

6.4.1 Butter powder. It has been found that Butter Powder B-1022 manufactured by Diehl Inc., meets the requirements of 3.2.2 and performs satisfactorily in this product.

6.4.2 Creaming agent, non-dairy. It has been found that San-A-Creme CWS 881 manufactured by Beatreme Foods, Inc., meets the requirements of 3.2.6 and performs satisfactorily in this product.

6.5 Subject term (key word) listing.

Canned food
Combat field feeding
Hospital
Jaw injury
Liquid diet
Operational ration
Pouch food
Shelf stable

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Custodians:

Army - GL
Navy - SA
Air Force - 50

Review activities:

Army - MD, QM
Navy - MC
DP - SS

Preparing activity:

Army - GL

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

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2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-S-44441	1992 September 30

3. DOCUMENT TITLE

SQUASH, BUTTERED, POWDER (FOR ORAL LIQUID FEEDING)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON/DSN
U.S. Army Natick RD&E Center	508-651-4501	256-4501
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: SATNC-WTP Natick, MA 01760-5018	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 NATICK OP4, JUL 91 Previous editions are obsolete. 198/29

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